

```
!pip install pandas
```

```
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (2.2.2)
Requirement already satisfied: numpy>=1.26.0 in /usr/local/lib/python3.12/dist-packages (from pandas) (2.0.2)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas) (2026.1)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas) (1.17.0)
```

```
import pandas as pd
print(pd.__version__)
```

```
2.2.2
```

```
import numpy as np
print(np.__version__)
```

```
2.0.2
```

series:1D array with heterogeneous datatype

```
s=pd.Series([1,2,3,4,5,6])
```

```
print(s)
```

```
0    1
1    2
2    3
3    4
4    5
5    6
dtype: int64
```

```
#custom index
s=pd.Series([1,2,3,4,5,6],index=['a','b','c','d','e','f'])
print(s)
```

```
a    1
b    2
c    3
d    4
e    5
f    6
dtype: int64
```

```
student={
    "name":"charan",
    "branch":"aiml"
}

se=pd.Series(student)
print(se)
```

```
name      charan
branch    aiml
dtype: object
```

```
n=pd.Series([10,20,30,40])
```

```
#n*2
#n+10
n%2
```

```

  0
0  0
1  0
2  0
3  0

dtype: int64
```

```
#statistics
n=pd.Series([12,20,30,40,50])
n.max()

50
```

```
n.min()

12
```

```
n.sum()

np.int64(152)
```

```
n.mean()

np.float64(30.4)
```

```
n.std()

15.192103211866355
```

```
n.describe()

      0
count  5.000000
mean   30.400000
std    15.192103
min    12.000000
25%    20.000000
50%    30.000000
75%    40.000000
max    50.000000

dtype: float64
```

```
n[n>20 ]

      0
2    30
3    40
4    50

dtype: int64
```

```
#vectorized string operations
names=pd.Series(['cherry','vinay','ashok','siva'])
names.str.lower()
```

```

      0
0  cherry
1  vinay
2  ashok
3  siva

dtype: object

```

```
names.str.len()
```

```

      0
0  6
1  5
2  5
3  4

dtype: int64

```

```
names.str.upper()
```

```

      0
0  CHERRY
1  VINAY
2  ASHOK
3  SIVA

dtype: object

```

```
names.str.startswith('s')
```

```

      0
0  False
1  False
2  False
3  True

dtype: bool

```

```

students={
    "names":["vamsi",'siva','charan','abu','ibu'],
    "branch":["aiml",'aiml','aiml','aiml','aiml']
}

s=pd.DataFrame(students)
print(s)

```

```

   names branch
0  vamsi  aiml
1   siva  aiml
2 charan  aiml
3    abu  aiml
4    ibu  aiml

```

```

#from list of dictionaries
std=[
    {"name":'vamsi',"branch":'aiml',"age":20},
    {"name":'sai',"branch":'aiml'},
    {"name":'vamsi',"branch":'aiml'}
]

df=pd.DataFrame(std)

```

```
print(df)
```

```
   name branch  age
0  vamsi   aiml  20.0
1    sai   aiml   NaN
2  vamsi   aiml   NaN
```

```
#from numpy array
arr=np.random.rand(3,4)
c=pd.DataFrame(arr,columns=['A','B','C','D'],index=['x','y','z'])
print(c)
```

```
   A         B         C         D
x  0.024840  0.021424  0.556442  0.421325
y  0.856488  0.030352  0.329924  0.384396
z  0.259809  0.909543  0.496620  0.557366
```

```
pd.DataFrame(arr,columns=['A','B','C','D'],index=['x','y','z'])
```

	A	B	C	D
x	0.024840	0.021424	0.556442	0.421325
y	0.856488	0.030352	0.329924	0.384396
z	0.259809	0.909543	0.496620	0.557366

```
#Exploring Data Frame
data={
    "name":["vamsi",'siva','charan','abu','ibu'],
    "age":[20,21,22,23,24],
    "score":[90,91,92,93,94],
    "grade":["A','B','A','A','A']
}
```

```
d=pd.DataFrame(data)
print(d)
```

```
   name  age  score grade
0  vamsi   20    90     A
1   siva   21    91     B
2 charan   22    92     A
3   abu   23    93     A
4   ibu   24    94     A
```

```
d.head(4)
```

	name	age	score	grade
0	vamsi	20	90	A
1	siva	21	91	B
2	charan	22	92	A
3	abu	23	93	A

Next steps: [Generate code with d](#) [New interactive sheet](#)

```
d.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5 entries, 0 to 4
Data columns (total 4 columns):
#   Column  Non-Null Count  Dtype
---  -
0    name    5 non-null        object
1    age      5 non-null        int64
2    score    5 non-null        int64
3    grade    5 non-null        object
dtypes: int64(2), object(2)
memory usage: 292.0+ bytes
```

```
d.describe()
```

	age	score
count	5.000000	5.000000
mean	22.000000	92.000000
std	1.581139	1.581139
min	20.000000	90.000000
25%	21.000000	91.000000
50%	22.000000	92.000000
75%	23.000000	93.000000
max	24.000000	94.000000

d.index

RangeIndex(start=0, stop=5, step=1)

d.dtypes

	0
name	object
age	int64
score	int64
grade	object

dtype: object

d.nunique()

	0
name	5
age	5
score	5
grade	2

dtype: int64

d.isnull().sum()

	0
name	0
age	0
score	0
grade	0

dtype: int64

d.columns

Index(['name', 'age', 'score', 'grade'], dtype='object')

d.name

	name
0	vamsi
1	siva
2	charan
3	abu
4	ibu

dtype: object

d[['name', 'age']]

	name	age
0	vamsi	20
1	siva	21
2	charan	22
3	abu	23
4	ibu	24

d.loc[0]

	0
name	vamsi
age	20
score	90
grade	A

dtype: object

d.loc[0:2]

	name	age	score	grade
0	vamsi	20	90	A
1	siva	21	91	B
2	charan	22	92	A

d.loc[0:2, 'name': 'score']

	name	age	score
0	vamsi	20	90
1	siva	21	91
2	charan	22	92

d.iloc[0]

	0
name	vamsi
age	20
score	90
grade	A

dtype: object

d.iloc[0:2]

	name	age	score	grade	
0	vamsi	20	90	A	
1	siva	21	91	B	

d.iloc[0:2,0:2]

	name	age	
0	vamsi	20	
1	siva	21	

d.iloc[-1]

	4
name	ibu
age	24
score	94
grade	A
dtype:	object

d.loc[0,'name']

'vamsi'

d.iloc[0,0]

'vamsi'

Start coding or [generate](#) with AI.